



Toy-Based Technologies for Children with Disabilities Simultaneously Supporting Self-Directed Mobility, Participation, and Function: A Tech Report

Samuel W. Logan^{1*}, Heather Ann Feldner², Kathleen R. Bogart¹, Brianna Goodwin¹, Samantha M. Ross¹, Michele Ann Catena¹, Austin A. Whitesell¹, Zachary J. Sefton¹, William D. Smart¹ and James Cole Galloway³

¹ Oregon State University, Corvallis, OR, USA, ² University of Washington, Seattle, WA, USA, ³ University of Delaware, Newark, DE, USA

OPEN ACCESS

Edited by:

Alessio Merola,
Magna Græcia University, Italy

Reviewed by:

Roslyn Watson Livingstone,
Sunny Hill Health Centre for Children,
Canada

Carole W. Dennis,
Ithaca College, USA

*Correspondence:

Samuel W. Logan
sam.logan@oregonstate.edu

Specialty section:

This article was submitted to
Biomedical Robotics,
a section of the journal
Frontiers in Robotics and AI

Received: 08 August 2016

Accepted: 09 February 2017

Published: 02 March 2017

Citation:

Logan SW, Feldner HA, Bogart KR,
Goodwin B, Ross SM, Catena MA,
Whitesell AA, Sefton ZJ, Smart WD
and Galloway JC (2017) Toy-Based
Technologies for
Children with Disabilities
Simultaneously Supporting
Self-Directed Mobility, Participation,
and Function: A Tech Report.
Front. Robot. AI 4:7.
doi: 10.3389/frobt.2017.00007

Go Baby Go is a community-based outreach, research, and clinical treatment program that works with families, clinicians, service providers, and industry to provide pediatric equipment to children with disabilities for movement, mobility, and socialization. The purpose of the current technical report is to describe two innovative and novel extensions of the initial modified ride-on car technology that encourages self-directed mobility, participation, and function for young children with disabilities. The current report provides a description of the “sit-to-stand” and “throw-baby-throw” ride-on car technologies: (1) *Sit-to-stand modified ride-on car technology*: commercially available, battery-operated, ride-on cars are adapted by installing a large, easy-to-press activation switch that is placed on the steering wheel. The switch has a large surface area and turns on at the slightest touch allowing easier activation for children with disabilities. Common materials such as PVC pipe, swimming kickboards and fun noodles, and Velcro are used to build a custom seating system that provides optimum support for each child. There are no commercially available powered wheelchairs for children with disabilities from birth to 3 years old. An innovative sit-to-stand version of modified ride-on car will be presented and requires a child to stand up in order to activate the switch to encourage the physical skills of pulling from sit-to-stand, weight bearing, and balance; (2) *Throw-baby-throw technology*: children with disabilities that experience limited upper-extremity function are not able to throw an object. The throw-baby-throw technology includes modification of a commercially available, toy-based ball pitching machine that projects foam balls. We modified the toy so that it is switch activated and attached to a modified ride-on car. Our modified throwing device allows children with disabilities to engage in the fundamental motor skill of throwing. Both technologies presented in this technical report embrace the holistic view of the International Classification of Functioning, Disability, and Health framework by placing equal emphasis on the body structure and function, activity, and



基于玩具的技术用于残障儿童，同时支持自主移动、参与和功能：技术报告

塞缪尔·W·洛根^{1*}，希瑟·安·费尔德纳²，凯瑟琳·R·博加特¹，布里安娜·古德温¹，萨曼莎·M·罗斯¹，米歇尔·安·卡特纳¹，奥斯汀·A·怀特塞尔¹，扎卡里·J·塞夫顿¹，威廉·D·斯马特¹和詹姆斯·科尔·加洛韦³

¹ 美国俄勒冈州立大学，科瓦利斯，² 美国华盛顿大学，西雅图，³ 美国特拉华大学，纽瓦克

OPEN ACCESS

编辑：
阿莱西奥·梅罗拉，意大利大希腊大学

审核人：
罗斯林·沃森·利文斯通，加拿大儿童阳光山健康中心 卡罗尔·W·丹尼斯，美国伊萨卡学院

*通信作者：Samuel
W. Logan sam.logan@oregonstate.edu

专科部分：
本文提交给《生物医学机器人学》，这是《机器人与人工智能前沿》期刊的一个分支栏目

收到：2016年8月08日 接受：
2017年2月09日 发布：2017年3月02日

引用：
Logan SW, Feldner HA, Bogart KR, Goodwin B, Ross SM, Catena MA, Whitesell AA, Sefton ZJ, Smart WD and Galloway JC (2017) 基于玩具的残障儿童技术，同时支持自主移动、参与和功能：一份技术报告. *Front. Robot. AI* 4:7. doi: 10.3389/frobt.2017.00007

Go Baby Go 是一个以社区为基础的外展、研究和临床治疗项目，与家庭、临床医生、服务提供者和行业合作，为有残疾的儿童提供用于运动、行动能力和社交的儿童设备。本技术报告的目的是描述最初的改装电动车技术的两个创新和新颖扩展，这些技术鼓励残疾儿童自我定向的移动、参与和功能。本报告提供了“坐到站”和“抛宝宝”电动车技术的描述：(1) 坐到站改装电动车技术：市面上可购买的电池驱动电动车通过在方向盘上安装一个大而易按的激活开关进行改装。该开关表面面积大，轻触即可启动，使残疾儿童更容易激活。

participation domains. Both technologies specifically target the advancement of physical skills while simultaneously providing opportunities to engage in experiences associated with activity and participation.

Keywords: mobility, children, do-it-yourself, medical technology devices, rehabilitation

INTRODUCTION

Self-directed mobility is a fundamental human right, especially for young children with disabilities who rely on adults to provide them access to mobility experiences. Self-directed mobility is defined as movement that is initiated by an individual and may include independent locomotion such as walking, the use of mobility technology such as gait trainers, standers, or the use of powered mobility devices including motorized wheelchairs, battery-operated ride-on toy cars, or other similar devices. Children without disabilities experience developmental gains in cognition, language, and social skills as a result of the onset of independent locomotion that occurs between 10 and 14 months of age (Campos et al., 2000; Uchiyama et al., 2008; Anderson et al., 2013).

Regardless of ability, the grounded cognition framework (aka embodied cognition or embodied development) (Thelen and Smith, 1994; Thelen et al., 2001; Smith and Gasser, 2005) suggests that learning occurs through movement as bodies and minds interact with and within environments and experiences of particular cultural or social contexts (Lobo et al., 2013). Research in developmental psychology and pediatric rehabilitation demonstrates that *active* (i.e., self-directed) movement experiences are fundamentally different and more advantageous compared to passive movement experiences (Held and Hein, 1963; Libertus and Needham, 2010; Jones et al., 2012; Lobo et al., 2013). Thus, the relatively passive lives of young children with disabilities who experience limited mobility are especially disturbing.

Over 30 years of research has demonstrated that young children with disabilities who experience the most severe delays in mobility can use powered mobility devices for self-directed mobility (Butler et al., 1983; Galloway et al., 2008; Lynch et al., 2009; Livingstone, 2010; Ragonesi et al., 2010; Jones et al., 2012). Equally important, using this technology results in some of the developmental gains associated with the onset of crawling and walking (Butler et al., 1983; Galloway et al., 2008; Lynch et al., 2009; Livingstone, 2010; Ragonesi et al., 2010; Jones et al., 2012). Unfortunately, the reality is that many if not most of the children who could benefit from powered mobility do not gain access to this technology until after 5 years of age (Nicholson and Bonsall, 2002). Given the importance of self-directed mobility, we believe that this lack of access is nothing less than a form of human rights violation—but a violation that is readily eliminated as the barriers to access and solutions are both known and being addressed by our work and others (Feldner et al., 2016).

One significant barrier that prevents widespread acceptance of access to self-directed mobility experiences is the absence of commercially available powered mobility devices that are specifically designed for young children with disabilities and accessible to all. For example, the Wizzybug® manufactured by designability® is

commercially available but is not available for purchase in North America. Another example is the Skippi© manufactured by Ottobock© but starts at a base price of \$10,000. Furthermore, the cost of other available powered mobility devices can be up to \$30,000 or more (Cooper et al., 2008). Insurance companies routinely deny funding requests based on cost, environmental accessibility, or a child's young age (Nicholson and Bonsall, 2002; Staincliffe, 2003). Modifying off-the-shelf, battery-operated ride-on cars has emerged as an alternative to traditional powered mobility devices (Huang and Galloway, 2012; Logan et al., 2014, 2016).

Go Baby Go is a community-based outreach, research, and clinical treatment program that works with families, clinicians, service providers, and industry to provide pediatric equipment to children with disabilities for movement, mobility, and socialization. The modified ride-on car technology, which was explored intermittently in the late 1980s and early 1990s, re-emerged from the Go Baby Go program in 2012 (Chiulli et al., 1988; Cooper et al., 1992). Commercially available, battery-operated, ride-on cars are adapted by installing a large, accessible activation switch. Common materials such as PVC pipe, swimming kickboards, and Velcro are used to build a customized seating system for support. Modified ride-on cars offer a low-cost, powered mobility device that is affordable, accessible, and fun. Because the device is a mainstream toy, caregivers and clinicians can address child and family goals without the social stigma of a medical device (Gibson et al., 2012). Not surprisingly, evidence is emerging that suggests positive developmental gains for children with disabilities who use modified ride-on cars for self-directed mobility (Huang and Galloway, 2012; Logan et al., 2014, 2016).

The purpose of the current technical report is to describe two innovative and novel extensions of the initial modified ride-on car technology that encourages self-directed mobility, participation, and function for young children with disabilities. The general features, modifications, and clinical applications of the modified ride-on car technology are described in a previously published technical report (Huang and Galloway, 2012). One major feature of the initial modified ride-on car technology is that the activation switch is pressed while a child is seated. A child must use their hand, trunk, or head in order to press the activation switch (see **Figure 1**). The current report extends that work to describe “sit-to-stand” and “throw-baby-throw” ride-on car technologies.

SIT-TO-STAND MODIFIED RIDE-ON CAR TECHNOLOGY

Scientific Background

Young children who experience mobility delays often experience delays across developmental domains (Schoenmakers et al., 2005;

参与领域。这两种技术特别针对身体技能的提升，同时提供参与与活动相关体验的机会。

关键词：移动性、儿童、自己动手、医疗技术设备、康复

介绍

自主移动是一项基本人权，特别是对于依赖成人提供移动体验的残疾幼儿而言。自主移动被定义为由个人自主发起的移动行为，可能包括独立行走、使用步行训练器、站立器等移动技术，或使用包括电动轮椅、充电玩具车或其他类似装置在内的动力移动设备。没有残疾的儿童在10至14个月之间独立移动的开始会带来认知、语言和社交技能的发展（Campos 等，2000；Uchiyama 等，2008；Anderson 等，2013）。

无论能力如何，扎根认知框架（又称具身认知或具身发展）（Thelen 和 Smith, 1994；Thelen 等，2001；Smith 和 Gasser, 2005）表明，学习是通过身体和思想在特定文化或社会环境与经验中互动而发生的运动过程（Lobo 等，2013）。发展心理学和儿童康复研究表明，积极的（即自我主导的）运动体验与被动运动体验存在根本区别，并且更有优势（Held 和 Hein, 1963；Libertus 和 Needham, 2010；Jones 等，2012；Lobo 等，2013）。因此，行动能力有限的残障儿童所经历的相对被动的生活尤为令人担忧。

超过30年的研究表明，经历最严重运动能力延迟的残疾幼儿可以使用动力辅助移动设备进行自主移动（Butler 等，1983；Galloway 等，2008；Lynch 等，2009；Livingstone, 2010；Ragonesi 等，2010；Jones 等，2012）。同样重要的是，使用这项技术会带来一些与开始爬行和行走相关的发展性收益（Butler 等，1983；Galloway 等，2008；Lynch 等，2009；Livingstone, 2010；Ragonesi 等，2010；Jones 等，2012）。不幸的是，现实情况是，许多甚至大多数本可以从动力辅助移动中受益的儿童，直到5岁以后才获得这种技术（Nicholson 和 Bonsall, 2002）。鉴于自主移动的重要性，我们认为这种缺乏获取的情况无异于一种人权侵犯。

阻碍广泛接受自主移动体验的一个重要障碍是缺乏专为残疾儿童设计且对所有人都可访问的商业化动力移动设备。例如，由 designability© 制造的 Wizzybug© 是

商业上可获得，但在北美不可购买。另一个例子是由 Ottobock© 生产的 Skippi©，起价为10,000美元。此外，其他可用的动力移动装置的费用可高达30,000美元或更多（Cooper 等，2008）。保险公司通常会基于成本、环境可及性或孩子年龄过小而拒绝资助请求（Nicholson 和 Bonsall, 2002；Staincliffe, 2003）。改装现成的电池驱动的儿童乘骑车已成为传统动力移动装置的替代方案（Huang 和 Galloway, 2012；Logan 等，2014, 2016）。

Go Baby Go 是一个以社区为基础的外展、研究和临床治疗项目，与家庭、临床医生、服务提供者和行业合作，为有残障的儿童提供用于运动、移动和社交的儿童设备。改装驾驶玩具车技术在 1980 年代末和 1990 年代初曾间歇性探索，并在 2012 年通过 Go Baby Go 项目重新出现（Chiulli 等，1988；Cooper 等，1992）。商业可获得的电池驱动驾驶玩具车通过安装一个大而易于操作的激活开关进行改装。常见材料如 PVC 管、游泳踢板和魔术贴被用来搭建定制化的座椅系统以提供支撑。改装驾驶玩具车提供了一种低成本的动力移动设备，既经济实惠，又易于使用，而且充满乐趣。由于该设备是一种主流玩具，照护者和临床医生可以在不带医疗设备社会污名的情况下，实现儿童和家庭的目标。

本技术报告的目的是描述对最初的改良型儿童电动车技术的两项创新性扩展，该技术旨在促进残障幼儿的自主移动、参与和功能发展。改良型儿童电动车技术的一般特征、改装方法及临床应用已在先前发表的技术报告中进行了描述（Huang 和 Galloway, 2012）。最初改良型儿童电动车技术的一个主要特点是儿童在座位上时按下启动开关。儿童必须使用手、躯干或头部来按下启动开关（见图 1）。本报告在此基础上进一步描述了“坐到站”和“抛宝宝”两种儿童电动车技术。

坐到站式改装骑乘车技术

科学背景

经历运动能力发育迟缓的幼儿通常在各个发展领域也会出现迟缓（Schoenmakers 等，2005；

Webster and Carter, 2007; Wuang et al., 2008; Carter et al., 2010). For example, young children who experience Down syndrome often will ultimately sit, stand, and walk up to a year or later than peers without disabilities (Tudella et al., 2011). These delays seriously restrict their interaction with family and friends, and their general development (Fidler, 2005). Ulrich and colleagues' pioneering work on treadmill training suggests that if training is started early, is meaningful to the infant, is systematically progressed, and is family centered, then physical skills, such as walking, of young children who experience Down syndrome can be advanced (Ulrich et al., 2001; Wu et al., 2010). For example, with less than 10 min of treadmill stepping per day for 6 months, young children who experience Down syndrome walked more than 100 days earlier than controls who received only standard care (Ulrich et al., 2001). Because treadmills are typically stationary and inside, children cannot use them to socialize through play with mobile peers in either indoor or outdoor environments. Sit-to-stand modified ride-on cars have the potential to be incorporated in daily experiences of children and provide opportunities to gain physical skills (balance, strength, coordination), functional skills (standing, sit-to-stand transition), and participation skills (use of car with peers for play).

The sit-to-stand technology is an innovative and novel treatment option that can be used in a variety of real-world settings. A multisite study of the impact of this technology will be completed within the year; however, previously published pilot work suggests that caregivers and clinicians should expect gains from daily use. For example, a recent case report found that a 4-year old that experiences a mobility disability engaged in more play behaviors with peers while using a sit-to-stand modified ride-on

car compared to using forearm crutches (Logan et al., 2017). This case is especially instructive as this child walked with crutches and would be considered very high functioning for a child who experiences mobility impairments. Yet, in important educational environments larger than the classroom—such as gymnasium and playground—his mobility and socialization were significantly limited and these limitations were lessened with use of a modified ride-on car. Group studies are underway and will provide a more detailed understanding of the physiological and psychosocial benefits of daily sit-to-stand use of modified ride-on cars.

Technical Background

The philosophy behind Go Baby Go technology is to provide safe yet speedy access to important mobility technology that can be purchased or built by every family and clinician planet wide. As such, the sit-to-stand technology includes modification of off-the-shelf, battery-operated, ride-on cars [see **Table 1** for a description of available 6 V, all-terrain vehicle (ATV) style ride-on cars]. Below we describe a common modification package.¹

The ATV style ride-on car typically comes with a pre-installed activation switch that is located on the handlebars and used for the seated mode. The switch contacts are normally open, which means that the switch has to be pressed in order to activate. *Modified drive system:* a SparkFun® Big Dome momentary push button (\$9.95; product ID: 11274) was installed on top of the seat and used for the standing mode. The switch circuit configuration is “normally closed,” which means that pressure on the switch has to be released in order to activate. Thus, a child must pull him or herself from sit-to-stand, thereby activating the switch (see **Figure 2** for description of switch wiring of seated and standing modes). Installation on top of the seat allows for easy adjustment

¹This modification description is for demonstration purposes. We suggest modifications be inspected by an engineering expert.

Original Seated version of the Go Baby Go technology



FIGURE 1 | Child uses a modified ride-on car for self-directed mobility, exploration, and play. The car is activated by pressing the center activation switch.

TABLE 1 | Description of available 6 V, all-terrain vehicle (ATV) style ride-on cars.

	Fisher-Price® Lil' Quad	Kid Trax®
Availability	Amazon, Toys R Us, Wal-Mart, other retailers	Target, Toys R Us, Wal-Mart, other retailers
Cost	\$65 and up	\$55 and up
Age, years	1–3	1.5+
Weight, lbs.	40	35–45
Speed, miles per hour	2	1.5
Seating capacity	1	1
Wheels	4	4
Terrain	Hard surfaces, grass	Hard surfaces, grass
Styles	Trains, ATVs, farm vehicles	ATVs
Designs/frame options	Colorful, cartoon characters with plastic frame	Colorful, cartoon characters with plastic frame
Battery/drive	6 V/single gearbox	6 V/single gearbox
Noise level	Minimally distracting	Minimally distracting
Activation	Push button	Push button
Transportation	Any type of car	Any type of car
Maintenance	Keep clean, regularly check battery power	Keep clean, keep dry, store away from stoves and heaters

Information gathered from manufacturers and/or distribution websites.

韦伯斯特和卡特, 2007; Wuang 等, 2008; 卡特等, 2010)。例如, 经历唐氏综合症的幼儿通常最终坐、站和走路的时间比没有残疾的同龄人晚一年或更久 (Tudella 等, 2011)。这些延迟严重限制了他们与家人和朋友的互动, 以及他们的整体发展 (Fidler, 2005)。Ulrich 及其同事在跑步机训练方面的开创性工作表明, 如果训练从早期开始, 对婴儿具有意义, 按系统进行进展, 并以家庭为中心, 那么经历唐氏综合症的幼儿的身体技能, 例如行走, 可以得到提高 (Ulrich 等, 2001; 吴等, 2010)。例如, 每天在跑步机上练习不到10分钟, 持续6个月, 经历唐氏综合症的幼儿比只接受常规护理的对照组提前走路超过100天 (Ulrich 等)。

坐-站技术是一种创新且新颖的治疗选择, 可在各种现实环境中使用。一项关于该技术影响的多中心研究将在一年内完成; 然而, 先前发表的试点研究表明, 护理人员和临床医生应预期通过日常使用获得收益。例如, 最近的一份病例报告发现, 一名患有行动障碍的四岁儿童在使用坐-站改装骑乘玩具时, 与同伴的游戏行为增多。

与使用前臂拐杖相比 (Logan 等, 2017)。这个案例尤其具有启发性, 因为这个孩子使用拐杖行走, 对于一名有移动障碍的孩子来说, 他的功能被认为是非常高的。然而, 在教室以外的重要教育环境中——例如体育馆和操场——他的移动能力和社交能力受到显著限制, 而使用改装的可坐骑车可以减轻这些限制。相关的群体研究正在进行中, 将提供关于每日使用改装可坐骑车进行坐-站训练的生理和心理社会益处的更详细理解。

技术背景

Go Baby Go 技术背后的理念是提供安全而快速的方式, 让全球每个家庭和临床医生都能够购买或自行制作重要的移动技术。因此, 坐到站立的技术包括对现成的电池驱动骑乘车的改装【参见表1, 了解可用的 6V 全地形车辆(ATV)风格骑乘车的描述】。下面我们描述一种常见的改装方案。

ATV 风格的可骑乘汽车通常配有预先安装的激活开关, 该开关位于车把上, 用于坐姿模式。开关触点通常是常开型, 这意味着必须按下开关才能激活。改装的驱动系统: 在座椅顶部安装了一个 SparkFun© Big Dome 瞬时按钮 (\$9.95; 产品编号: 11274), 用于站立模式。开关电路配置为“常闭”, 也就是说必须释放开关的压力才能激活。因此, 孩子必须从坐姿拉起自己到站姿, 从而激活开关 (有关坐姿和站姿模式开关接线的说明, 请参见图 2)。安装在座椅顶部便于调节。

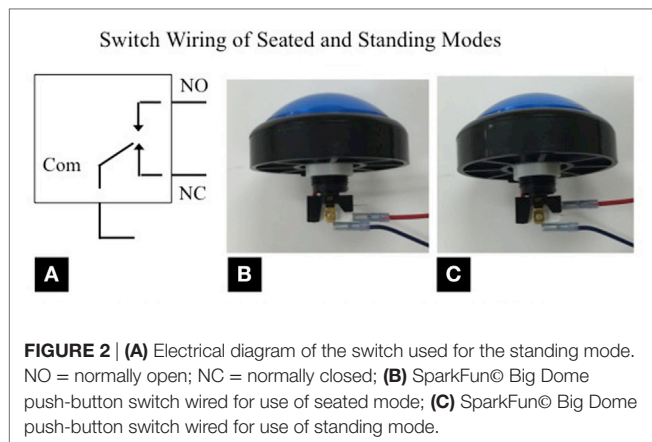
¹This modification description is for demonstration purposes. We suggest modifications be inspected by an engineering expert.

表 1 | 可用 6 V 全地形车 (ATV) 风格儿童电动车的描述。

	Fisher-Price© Lil' Quad	Kid Trax©
Availability	Amazon, Toys R Us, Wal-Mart, other retailers	Target, Toys R Us, Wal-Mart, other retailers
Cost	\$65 and up	\$55 and up
Age, years	1–3	1.5+
Weight, lbs.	40	35–45
Speed, miles per hour	2	1.5
Seating capacity	1	1
Wheels	4	4
Terrain	Hard surfaces, grass	Hard surfaces, grass
Styles	Trains, ATVs, farm vehicles	ATVs
Designs/frame options	Colorful, cartoon characters with plastic frame	Colorful, cartoon characters with plastic frame
Battery/drive	6 V/single gearbox	6 V/single gearbox
Noise level	Minimally distracting	Minimally distracting
Activation	Push button	Push button
Transportation	Any type of car	Any type of car
Maintenance	Keep clean, regularly check battery power	Keep clean, keep dry, store away from stoves and heaters

从制造商和/或分销网站收集的信息。





of switch placement along the length of the seat. This is important because the switch location can be moved as a child grows or if multiple children use the modified ride-on car in a clinical setting (see **Figure 3** for modified sit-to stand ride-on car).

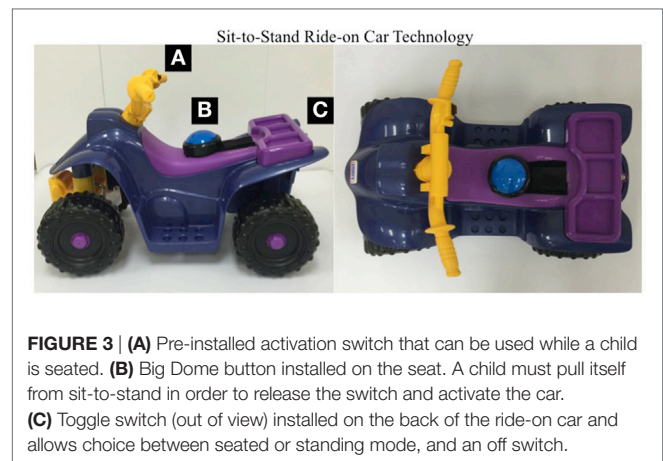
A major feature of the electrical modification is installation of a “double-pole, double-throw” (DPDT) toggle switch. The seated and standing mode switches are interfaced with the DPDT toggle switch. This allows for the modified ride-on car to be used in either the seated or standing mode. The DPDT toggle switch also provides an “off” setting that prevents unwanted use of the modified ride-on car, including for safe transfers in and out of the ride-on car (see **Figure 4** for description of toggle switch installation).

Seating and support system: the seating and support system for the sit-to-stand technology is very similar to the seated car versions (Huang and Galloway, 2012). However, different types of modifications are needed that provide additional support while a child is in the standing position. Low-cost and readily accessible PVC pipe and connectors were used to fabricate back and lateral support. Then, industrial strength Velcro was used to provide lateral support (see **Figure 5** for seating and support system examples).

THROW-BABY-THROW TECHNOLOGY

Scientific Background

An individual’s overall capability to perform a variety of skills and tasks related to the motor domain (aka motor competence) is related to positive developmental trajectories of health including physical activity, health-related fitness, perceived competence, and weight status (Robinson et al., 2015). One aspect of motor competence includes fundamental motor skills. Fundamental motor skills include object control/ball skills (i.e., throwing, catching, etc.), locomotor skills (i.e., running, jumping, etc.), and balance/stability skills (i.e., one-leg balance, etc.) (Clark and Metcalfe, 2002; Haywood and Getchell, 2009). These skills are considered the “building blocks” of more advanced, complex movements required to participate in sports, games, or other context specific physical activity. Fundamental motor skills, including throwing,



are not thought to develop naturally during childhood, but through high-quality teaching, practice, and experience (Logan et al., 2011). Children with disabilities may experience physical or cognitive differences that make it challenging to learn, practice, and develop the skill of throwing; yet deserve equitable access to participate in activities that require throwing to support positive development. Like with ride-on cars, simple modifications can improve the access to these important skills and in turn may continue to encourage positive peer interactions.

The skill of throwing may be modified to provide children with disabilities with opportunities to engage in physical activity similar to children without disabilities. For example, a larger object may be used so that a child is more easily able to grasp and throw. The goal of skill modification is that all children, regardless of ability, can be successful and active participants. However, the modification of a larger ball size, or other modifications, may not be suitable for young children with disabilities who experience severe limitations of arm movement. Thus, fewer strategies exist to modify the skill of throwing for young children with disabilities. This is likely due to their smaller body size, lack of muscle strength and coordination, and greater difficulty in relaying task instructions. These children may experience limited physical movement and are often left out of throwing activities or they are dependent upon adults to help them throw. The throw-baby-throw technology is an innovative and novel approach to encourage the physical skill of reaching, while allowing a child to engage in the fundamental motor skill of throwing.

Technical Background

Throw-baby-throw technology includes modification of off-the-shelf, battery-operated, toy-based ball pitching machines. For the current report, we modified the Fisher-Price® Triple Hit Baseball model (\$33.00, Product B6312). This model requires four D-cell batteries and operates by pressing an activation button on the device that launches four foam balls on a timer. This model uses an internal, spinning wheel (servomotor) to propel foam balls and includes an on/off switch that closes and opens the circuit, respectively. The servomotor rotates and launches the foam ball when the activation switch is pressed. After the launch, the spring

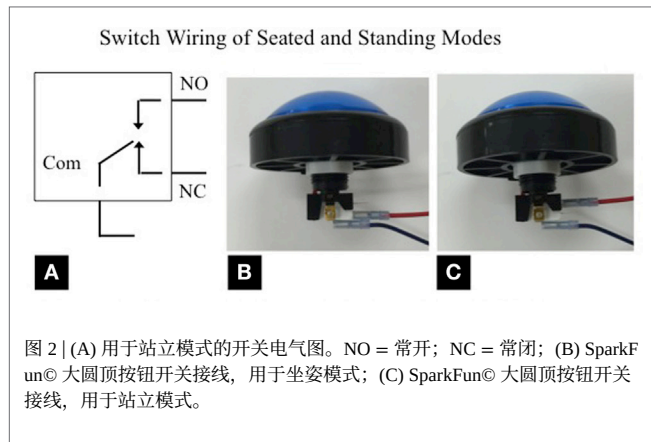


图2 | (A) 用于站立模式的开关电气图。NO = 常开; NC = 常闭; (B) SparkFun® 大圆顶按钮开关接线, 用于坐姿模式; (C) SparkFun® 大圆顶按钮开关接线, 用于站立模式。

沿座椅长度的开关放置。这很重要, 因为随着孩子的成长, 或者如果多个孩子在临床环境中使用改装的乘坐式电动车, 开关位置可以移动 (见图3 改装坐到站乘坐式电动车)。

电气改装的一个主要特点是安装了“双刀双掷” (DPDT) 拨动开关。坐姿和站姿模式开关与DPDT拨动开关相连接。这使得改装后的儿童电动车可以在坐姿或站姿模式下使用。DPDT拨动开关还提供“关”设置, 可防止儿童电动车被非意图使用, 包括在安全上下车时 (见图4了解拨动开关的安装说明)。

座椅和支撑系统: 坐-站技术的座椅和支撑系统与汽车座椅版本非常相似 (Huang 和 Galloway, 2012)。然而, 需要不同类型的改装, 以在儿童处于站立位置时提供额外的支撑。使用低成本且容易获取的PVC管和连接件制作背部和侧面支撑。然后, 使用工业强度的魔术贴提供侧面支撑 (座椅和支撑系统示例见图5)。

抛-宝宝-抛 技术

科学背景

个体在运动领域相关的各种技能和任务的整体能力 (也称为运动能力) 与健康的积极发展轨迹相关, 包括体育活动、健康相关的体适能、感知能力和体重状况 (Robinson 等, 2015)。运动能力的一个方面包括基本运动技能。基本运动技能包括物体控制/球类技能 (即投掷、接球等)、移动技能 (即跑步、跳跃等) 以及平衡/稳定技能 (即单腿平衡等) (Clark 和 Metcalfe, 2002; Haywood 和 Getchell, 2009)。这些技能被视为参与运动、游戏或其他特定情境体育活动所需的更高级、复杂动作的“构建模块”。基本运动技能, 包括投掷,

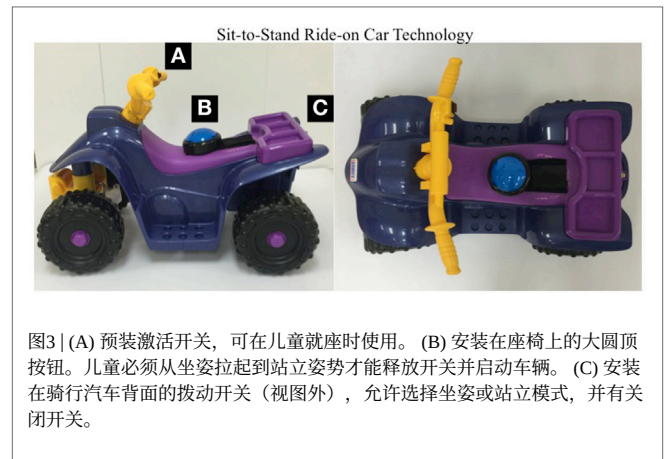


图3 | (A) 预装激活开关, 可在儿童就座时使用。 (B) 安装在座椅上的大圆顶按钮。儿童必须从坐姿拉起到站立姿势才能释放开关并启动车辆。 (C) 安装在骑行汽车背面的拨动开关 (视图外), 允许选择坐姿或站立模式, 并有关闭开关。

人们认为这些技能在童年期间不会自然发展, 而是通过高质量的教学、练习和经验 (Logan 等, 2011) 获得的。残疾儿童可能会经历身体或认知上的差异, 这使得学习、练习和发展投掷技能具有挑战性; 但他们同样应享有公平的机会参与需要投掷的活动, 以支持积极的发展。就像骑行玩具车一样, 简单的调整可以改善这些重要技能的获取, 从而可能继续促进积极的同伴互动。

投掷技能可以进行调整, 以便为有残疾的儿童提供与没有残疾的儿童类似的体育活动机会。例如, 可以使用较大的物体, 以便儿童更容易抓住和投掷。技能调整的目标是让所有儿童, 无论能力如何, 都能成为成功和积极的参与者。然而, 增大球的尺寸或其他调整可能不适合那些手臂活动能力受限严重的残疾幼儿。因此, 对于残疾幼儿来说, 调整投掷技能的方法较少。这很可能是由于他们身体较小、缺乏肌肉力量和协调性, 以及更难理解任务指令。这些儿童可能活动受限, 且常常被排除在投掷活动之外, 或者需要依赖成人的帮助才能进行投掷。

技术背景

投宝宝技术包括改装现成的、电池驱动的、以玩具为基础的投球机。在本次报告中, 我们改装了费雪® Triple Hit Baseball 型号 (\$33.00, 产品编号 B6312)。该型号需要四节D型电池, 通过按下设备上的激活按钮操作, 该按钮会以定时方式发射四个泡沫球。该型号使用内部旋转轮 (伺服电机) 来推动泡沫球, 并包括一个开/关开关, 分别用于闭合和打开电路。当按下激活开关时, 伺服电机旋转并发射泡沫球。发射后, 弹簧

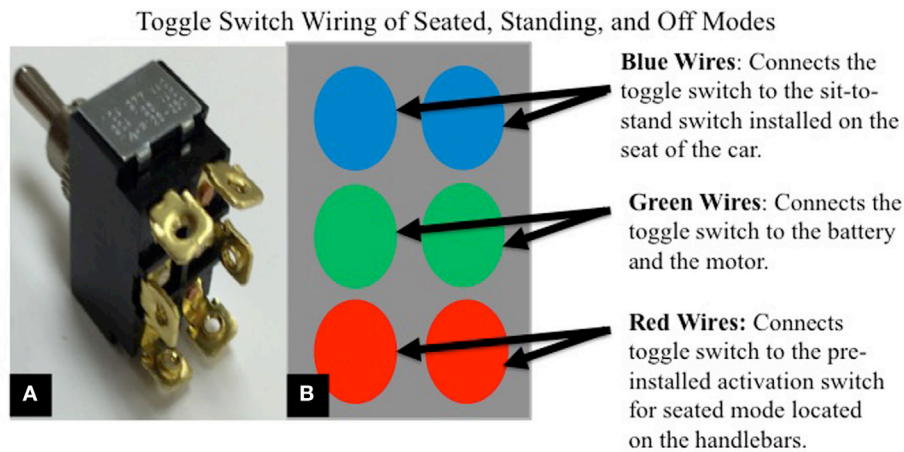


FIGURE 4 | (A) Double-pole double-throw toggle switch; **(B)** electrical diagram that outlines the interface between the toggle switch and the ride-on car. Note: the color of the wires may vary between ride-on car models. The colors displayed were chosen arbitrarily.

Seating and Support Systems of Sit-to-Stand Ride-on Car



FIGURE 5 | Three examples of seating and support systems that use PVC pipe and connectors, swimming kickboards, pool noodles, and Velcro.

inside the servomotor returns to its initial position (see **Table 2** for a description of the Fisher-Price® Triple Hit Baseball toy-pitching machine).

Modified Launch System

An Adafruit® large arcade momentary push button (\$9.95, Product 1185) was installed to allow for control of the foam ball launch. This allows for easier activation due to the large surface area of the arcade button and can be placed in a location that is accessible to each child. A potentiometer (i.e., variable resistor) from Digikey (\$5.34, Product CT2150-ND) was installed to allow

adjustment of the launch speed of the foam ball. The potentiometer is connected in-line with the motor in order to vary the voltage to it, thus varying its maximum speed (see **Figure 6** for description of electrical modifications). *Mounting system:* the Fisher-Price® Triple Hit Baseball model is attached to the side of the PVC seating and support system of the modified ride-on car *via* a wood panel and U-bolts. The installed large arcade button and potentiometer are attached to the top of the PVC seating and support system of the modified ride-on car *via* a separate wood panel and U-bolts (see **Figure 7** for a modified ride-on car that includes the throw-baby-throw technology).

Toggle Switch Wiring of Seated, Standing, and Off Modes

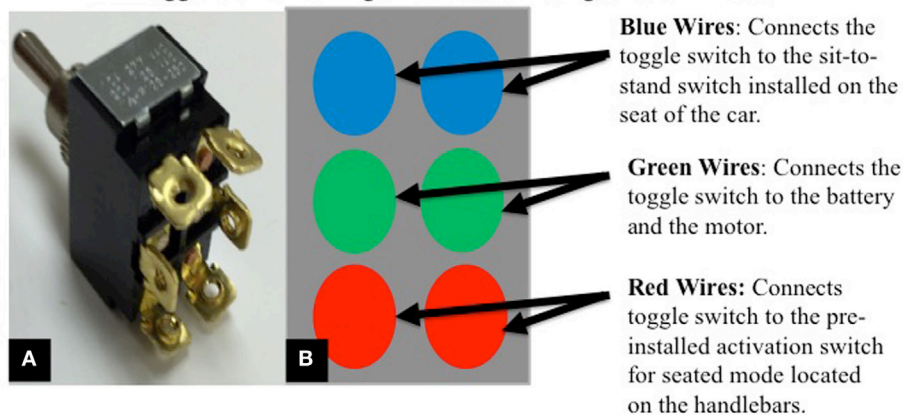


图4 | (A) 双刀双掷拨动开关; (B) 电气图, 概述拨动开关与电动玩具车之间的接口。注: 电线的颜色可能因儿童电动车型号而异。显示的颜色是随意选择的。

Seating and Support Systems of Sit-to-Stand Ride-on Car



图5 | 三个例子

使用PVC管和连接件的座椅和支撑系统, 游泳

踢板、泳池浮筒和魔术贴。

伺服电机内部返回到其初始位置 (有关 Fisher-Price® 三击棒球玩具投球机的说明, 请参见表 2)。

改进的发射系统

安装了一个 Adafruit® 大型街机瞬时按钮 (\$9.95, 产品编号 1185), 用于控制泡沫球发射。这由于街机按钮的大表面使得操作更加容易, 并且可以放置在每个孩子都能触及的位置。安装了一个来自 Digikey 的电位器 (即可变电阻) (\$5.34, 产品编号 CT2150-ND) 以允许

泡沫球发射速度的调节。电位器与电机串联连接, 以便改变电机的电压, 从而改变其最大速度 (电气改装说明见图 6)。安装系统: Fisher-Price® 三击棒球模型通过木板和 U 型螺栓固定在改装电动车的 PVC 座椅和支撑系统侧面。安装的大型街机按钮和电位器通过另一块木板和 U 型螺栓固定在改装电动车的 PVC 座椅和支撑系统顶部 (改装电动车及投掷宝宝技术见图 7)。

TABLE 2 | Description of the Fisher-Price® Triple Hit Baseball toy-pitching machine.

	Fisher-Price® Triple Hit Baseball
Availability	Amazon, Toys R Us, Wal-Mart, other retailers
Cost	\$20 and up
Age, years	3–7 ^a
Weight, lbs.	4.4
Speed, miles per hour	10.5 ^b
Battery/motor	6.3 V ^b /16.3 W ^b
Noise level	Difficult to talk over
Activation	On/off switch and a launch button
Transportation	Any car available
Maintenance	Keep clean, keep dry

Information gathered from manufacturers and/or distribution websites.

^aAccording to the manufacturer, this toy is not suitable for ages under 3 years. It contains one or more of the following items: marbles, small ball, or small parts. This recommendation is based upon the assumption that children will use product as a pitching machine.

^bMeasured values.

LIMITATIONS AND ADVANTAGES OF BOTH TECHNOLOGIES

One limitation includes that modifications range from relatively simple (sit-to-stand technology) to complex (throw-baby-throw technology). It may be difficult for individuals without prior engineering or electrical experience to complete the modifications on their own. Another limitation is the level of noise that ranges from minimally (sit-to-stand technology) to potentially distracting (throw-baby-throw technology). Finally, while families and volunteers can complete the modifications using step-by-step instructions, many people are hesitant to attempt the modifications themselves, citing intimidation, or a lack of perceived confidence in their technical abilities.

A limitation of both technologies is that often, a caregiver still may be required to provide assistance with steering (sit-to-stand technology) or helping to retrieve the foam balls (throw-baby-throw technology). However, it is hoped that incorporating the throwing device on to the car will also provide increased opportunities for peer-to-peer interaction in all aspects of ball game play, where a peer might engage in ball retrieval rather than a caregiver. In addition, children with disabilities may not have the coordination and bilateral upper-extremity function to simultaneously use both technologies at the same time. Children with disabilities may need to stop the ride-on car before activating the switch for throwing. However, the skill level required to run and throw at the same time is often beyond the capabilities of young children without disabilities. The throw-baby-throw technology is designed to emphasize social interaction, rather than proficiency of sport. Future research will examine the use of both technologies by children with disabilities in natural contexts such as early learning centers and within the home.

There are several advantages to the sit-to-stand and throw-baby-throw technologies. One advantage is that both technologies were developed using developmentally inspired principles of powered mobility technology (Feldner et al., 2016). These principles include: adaptive, flexible, durable, low-cost, esthetic,

and accessible [see Feldner et al. (2016) for a full description of each principle]. Another advantage is that both technologies are designed to be simultaneously “assistive” and “rehabilitative.” For example, a wheelchair is an assistive device that supports participation, but typically does not contribute to increasing function. By contrast, a treadmill is a rehabilitative device that supports function, but typically does not contribute to increasing participation. The sit-to-stand technology is assistive as it provides a means of self-directed mobility. This technology is also rehabilitative because embedded within its use is encouragement of the physical skills of pulling from sit-to-stand and dynamic-assisted standing and balance. The throw-baby-throw technology is assistive because it provides a means of throwing. This technology is also rehabilitative because embedded within its use is encouragement of the physical skill of reaching. The activation switch can be moved closer or further away, or on the left/right side or in midline in order to encourage reaching in a way that promotes physical strength and coordination.

Another advantage is that both technologies are lightweight and portable which promotes use within a variety of real-world contexts, such as the home, early childcare center, playground, and other community spaces. Due to this design feature, the technologies may be used much more frequently and for longer durations than traditional pediatric technology. Another advantage is the relatively low-cost of modification supplies that will enable greater access to all families. Finally, these toy-based technologies are inclusive and socially welcoming for children of all abilities, promoting further integration and participation of children with and without disabilities within the same play contexts.

Another advantage of the throw-baby throw technology is that it may foster independence. A majority of switch adapted toys focus on sensory consequences such as lights, sounds, and vibrations, rather than an emphasis on participation that may be provided by the throw-baby-throw technology. Typically, switch adapted toys are simply placed next to a child with a disability. This does not allow a child with a disability to move within the environment to initiate social and play interactions with others. The attachment of the throw-baby-throw technology to a modified ride-on car provides the potential of more independence in participating with others by allowing a child with a disability to make their own decisions with regard to exploring the environment and engaging in the skill of throwing.

DISCUSSION

Over the past 15 years, pediatric rehabilitation professionals have widely adopted a modern health framework, the World Health Organization's International Classification of Functioning, Disability, and Health (ICF), as a model of holistic practice (World Health Organization, 2001). The ICF includes three health domains: (1) body structure (anatomical parts of the body) and function (physiological functions of body systems); (2) activity (execution of a task or action by an individual); and (3) participation (involvement in life situations). The ICF also considers environmental factors that make up the physical, social,

表 2 | Fisher-price© 三重击棒球玩具—投球机的描述。

Fisher-Price© Triple Hit Baseball	
Availability	Amazon, Toys R Us, Wal-Mart, other retailers
Cost	\$20 and up
Age, years	3–7 ^a
Weight, lbs.	4.4
Speed, miles per hour	10.5 ^b
Battery/motor	6.3 V ^b /16.3 W ^b
Noise level	Difficult to talk over
Activation	On/off switch and a launch button
Transportation	Any car available
Maintenance	Keep clean, keep dry

信息来自制造商和/或分销网站。^a根据制造商的说法，此玩具不适合3岁以下儿童。
。它包含以下一个或多个物品：弹珠、小球或小零件。此建议基于假设儿童将产品用作投球机。^b测量值。

两种技术的限制与优势

一个限制包括改装的复杂程度从相对简单（坐到站技术）到复杂（扔宝宝技术）不等。对于没有工程或电气经验的个人来说，独自完成这些改装可能比较困难。另一个限制是噪音水平，从最小化（坐到站技术）到可能令人分心（扔宝宝技术）不等。最后，虽然家庭成员和志愿者可以按照逐步说明完成改装，但许多人对于自己尝试改装仍持犹豫态度，原因包括受到威吓或对自己的技术能力缺乏信心。

这两种技术的一个限制是，通常仍可能需要护理人员在操控方向（坐-站技术）或帮助捡回泡沫球（投宝宝-投技术）方面提供协助。然而，人们希望将投掷装置整合到汽车上，也能在球类游戏的各个方面提供更多同伴之间互动的机会，在这种情况下，同伴可能会参与捡球，而不是由护理人员来完成。此外，残障儿童可能没有足够的协调能力和双上肢功能，同时使用这两种技术。残障儿童可能需要在激活投掷开关之前先停下电动小车。然而，同时跑动和投掷所需的技能水平，往往超出了健康年幼儿童的能力范围。投宝宝-投技术…

坐-站和扔-婴儿-扔技术有几个优势。其中一个优势是，两种技术都是基于发展灵感的动力移动技术原则开发的（Feldner 等人, 2016）。这些原则包括：适应性、灵活性、耐用性、低成本、美观，

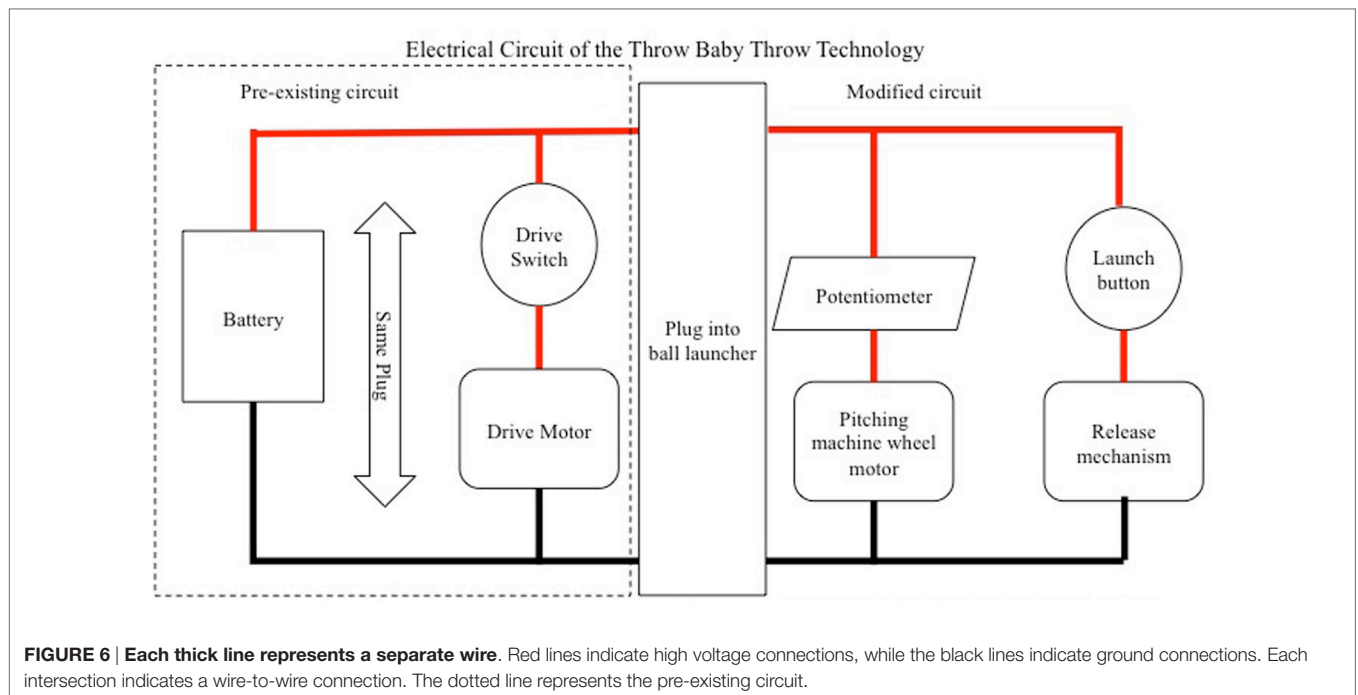
并且可访问 [详见 Feldner 等人 (2016) 关于每个原则的完整描述]。另一个优点是这两种技术都被设计为同时具有“辅助性”和“康复性”。例如，轮椅是一种支持参与的辅助设备，但通常不会有助于功能的提升。相比之下，跑步机是一种支持功能的康复设备，但通常不会有助于增加参与度。坐到站技术是辅助性的，因为它提供了一种自我导向移动的手段。这项技术也是康复性的，因为在使用过程中，它能够鼓励拉动自身从坐到站以及动态辅助的站立和平衡的身体技能。投掷婴儿技术是辅助性的，因为它提供了一种投掷的手段。这项技术也是康复性的，因为在使用过程中，它包含了对…的支持。

另一个优势是这两种技术都是轻便和便携的，这促进了它们在各种现实环境中的使用，例如家庭、早期托儿中心、游乐场和其他社区空间。由于这一设计特性，这些技术的使用频率可能比传统儿童技术更高，使用时间也更长。另一个优势是修改材料的成本相对较低，这将使所有家庭更容易获得。最后，这些基于玩具的技术具有包容性，对各种能力的儿童都友好，促进了有障碍和无障碍儿童在相同游戏环境中的进一步融合和参与。

投掷婴儿投掷技术的另一个优点是它可能促进独立性。大多数改装开关玩具侧重于感官反馈，例如灯光、声音和振动，而不是强调通过投掷婴儿投掷技术提供的参与感。通常，改装开关玩具只是简单地放在有残疾的孩子旁边。这并不允许有残疾的孩子在环境中移动以主动与他人进行社交和游戏互动。将投掷婴儿投掷技术安装在改装的骑行玩具车上，可以通过让有残疾的孩子自主决定探索环境和进行投掷技能的方式，提高他们与他人互动的独立性潜力。

讨论

在过去的15年里，儿童康复专业人员广泛采用了一种现代健康框架——世界卫生组织的《国际功能、残疾及健康分类》(ICF)——作为整体实践的模式（世界卫生组织，2001）。ICF包括三个健康领域：(1) 身体结构（身体的解剖部位）和功能（身体系统的生理功能）；(2) 活动（个体执行任务或行为）；以及(3) 参与（参与生活情境）。ICF还考虑构成物理和社会环境的环境因素。



and attitudinal environment in which people live and conduct their lives (World Health Organization, 2001).

The ICF was generated out of concern that previous classifications of the various domains of health and functioning neglected to recognize the social and environmental barriers to function that exist external to an individual, thus situating disability as solely an individual problem (Wiaert and Darrah, 2002; Ostensjö et al., 2005; Gibson and Teachman, 2012). The ICF model, as a response to these concerns, contains body function and structure domains as well as activity and participation domains, all of which can influence or be influenced by personal factors and environmental contexts, but which are not meant to be linear or to imply causal relationships (Law et al., 2007; Gibson et al., 2009; Gibson and Teachman, 2012).

Despite the adoption of the ICF, the current culture of pediatric rehabilitation and caregivers alike view the use of powered mobility devices as a “last resort,” once all other efforts toward the skill of walking have been exhausted (Bottos and Gericke, 2003; Wiaert et al., 2004; Rosen et al., 2009; Feldner et al., 2016). This represents a dichotomous view of thinking suggesting that self-directed mobility should be achieved by *either* a powered mobility device *or* walking. This dichotomous view may persist, in part, because researchers and caregivers have observed that the inability to walk is considered the most visible signifier of disability (Wiaert et al., 2004; Daudji et al., 2011; Gibson and Teachman, 2012). For example, when comparing a walker device to a wheeled mobility device, a caregiver expresses

Especially the kids at school, they think it's pretty neat that he's walking like they are. He seems to be more of a boy than more of a, ah, than you know someone with a disability (McKeever et al., 2013, p. 386).

This view reinforces traditional emphasis on body structure and function goals of the ICF, which focuses on walking as “normalcy,” undermining other forms of mobility, and implying that participation is a lesser goal. Moreover, it equates disability with being *less than* a boy, or, to put it bluntly, *less than* human. Emerging research indicates that this sentiment is the consequence, in part, of a lack of positive disability narratives that challenge walking as the only means of “normalcy,” which, left unmitigated, often results in children with disabilities’ internalization of stigma and low self-worth (Gibson et al., 2011; Gibson and Teachman, 2012). Several studies cite the role of rehabilitation professionals as key in either maintaining or challenging these established normative views on disability and walking (Wiaert and Darrah, 2002; Bottos and Gericke, 2003; Wiaert et al., 2004; Gibson et al., 2011; Gibson and Teachman, 2012).

The sit-to-stand and throw-baby-throw technologies presented in this technical report embrace the holistic view of the ICF by placing equal emphasis on the body structure and function, activity, and participation domains. The sit-to-stand technology specifically targets the physical skills of pulling from sit-to-stand and assisted standing, skills that contribute to the onset of independent locomotion. The throw-baby-throw technology specifically targets the physical skill of reaching. These aspects of the technologies emphasize the body structure and function domain by encouraging the advancement of specific physical skills.

The sit-to-stand and throw-baby-throw technologies presented in this technical report simultaneously embrace the medical and social models of disability. The ICF represents an attempt to find balance between dominant disability models. In certain academic or advocacy circles, there is a broader, on-going societal conversation regarding different ways of thinking about

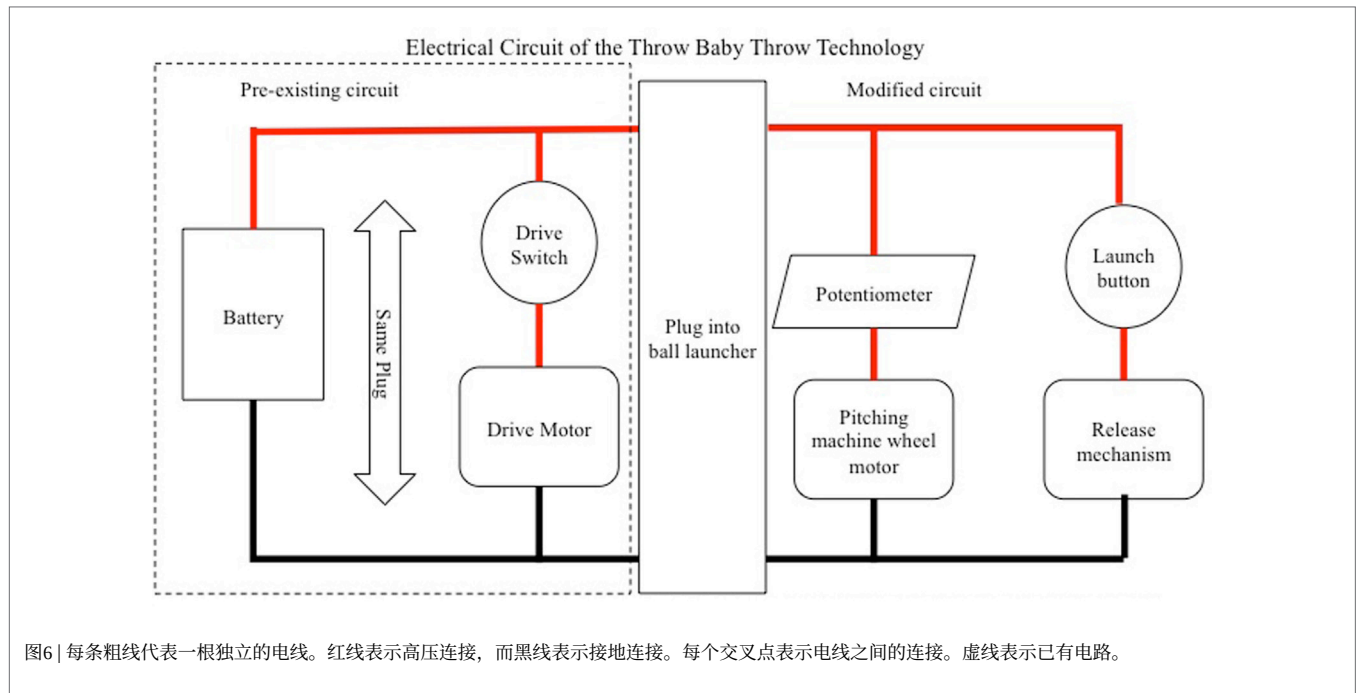


图6 | 每条粗线代表一根独立的电线。红线表示高压连接，而黑线表示接地连接。每个交叉点表示电线之间的连接。虚线表示已有电路。

以及人们生活和开展日常生活的态度环境（世界卫生组织，2001）。

国际功能、残疾和健康分类（ICF）的产生是出于对以往关于健康和功能各领域的分类未能认识到存在于个人之外的社会和环境障碍的担忧，因此将残疾仅仅定位为个人问题（Wiaert 和 Darrah, 2002; Ostensjø 等, 2005; Gibson 和 Teachman, 2012）。作为对这些问题的回应，ICF 模型包含身体功能和结构领域以及活动和参与领域，所有这些都可以影响或受个人因素和环境情境的影响，但并不意味着是线性的或暗示因果关系（Law 等, 2007; Gibson 等, 2009; Gibson 和 Teachman, 2012）。

尽管采用了国际功能、残疾与健康分类（ICF），当前儿科康复和照护者的文化仍普遍将使用动力移动设备视为“最后的手段”，一旦所有其他行走技能的努力都已耗尽（Bottos 和 Gericke, 2003; Wiaert 等, 2004; Rosen 等, 2009; Feldner 等, 2016）。这代表了一种二分思维模式，认为自我主导的移动能力应该通过动力移动设备或步行来实现。这种二分观念部分上可能得以持续，因为研究人员和照护者观察到，不能行走被认为是残疾最明显的标志（Wiaert 等, 2004; Daudji 等, 2011; Gibson 和 Teachman, 2012）。例如，当比较助步器和轮式移动设备时，一位照护者表达了

尤其是学校里的孩子们，他们觉得他像他们一样行走非常酷。他似乎更像一个普通的男孩，而不是，一个，你知道的，有残疾的人（McKeever 等, 2013, 第386页）。

这一观点强化了ICF对身体结构和功能目标的传统强调，即将行走视为“正常”，削弱了其他形式的移动能力，并暗示参与是次要目标。此外，它将残疾等同于不如男孩，或者坦率地说，不如人类。新兴研究表明，这种观念部分是由于缺乏积极的残疾叙事来挑战将行走作为唯一“正常”手段的观念，如果不加以改变，往往会导致残疾儿童内化污名感和低自尊（Gibson 等, 2011; Gibson 和 Teachman, 2012）。一些研究指出，康复专业人员在维持或挑战关于残疾和行走的这些既定规范观点中起着关键作用（Wiaert 和 Darrah, 2002; Bottos 和 Gericke, 2003; Wiaert 等, 2004; Gibson 等, 2011; Gibson 和 Teachman, 2012）。

本技术报告中提出的“坐到站”和“抛宝宝”技术采用了国际功能、残疾和健康分类（ICF）的整体视角，同等重视身体结构和功能、活动及参与领域。坐到站技术专门针对从坐到站的拉动及辅助站立的身体技能，这些技能有助于独立行动能力的开始。抛宝宝技术专门针对伸手的身体技能。这些技术方面通过促进特定身体技能的发展，强调了身体结构和功能领域。

本技术报告中提出的“坐到站”和“抛宝宝”技术同时采用了医学模型和社会模型的残疾观。国际功能、残疾与健康分类（ICF）代表了在主流残疾模型之间寻找平衡的一种尝试。在某些学术或倡导圈子中，关于不同思维方式的社会讨论正在更广泛地持续进行。

Throw Baby Throw Ride-on Car Technology

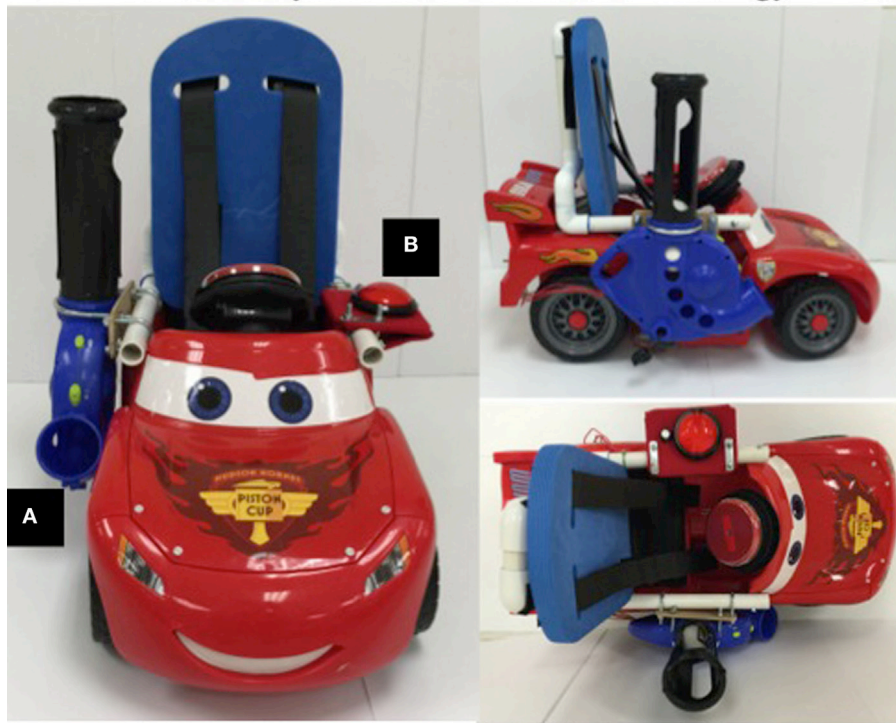


FIGURE 7 | (A) Fisher-Price® Triple Hit Baseball toy-pitching machine; **(B)** Adafruit® arcade button and potentiometer installed on a custom wood panel mounting system.

disability. However, knowledge of disability models themselves, and their implications, has yet to be translated into most of the lay-public's daily community life. Disability models are sets of beliefs about the cause, nature, and treatment of disability. The medical model views disability as an individual pathology that is determined through diagnostic criteria based on deviation from a statistically determined norm for the purpose of treatment or cure (Olkin, 2002). An alternative view is the social model of disability. In this model, the "problem" of disability is primarily socially constructed and society's lack of accommodation for people's differences is the cause of the disabling experience (Olkin, 2002). Therefore, while impairment may exist at an individual level, disability occurs because societies are structured under the assumption that everyone is healthy, without disability, and fits a culturally constructed norm or ideal (Wendell, 1996). The social model encourages the formation of an identity inclusive of disability as a positive experience, akin to a minority group, one that can help to improve an individual's quality of life and adaptability (Dunn and Burcaw, 2013; Bogart, 2014).

Both technologies presented in this technical report emphasize impairments at the individual level by advancing physical skills. These aspects of the technologies align more closely with the medical model. However, encouragement of physical skills is embedded within ride-on cars, a daily play, and exploration and participation experience of millions of children without

disabilities. Both technologies provide a means for children with disabilities to engage in similar childhood experiences. The low-cost, low-tech, and do-it-yourself aspects of the technologies promotes increased access for children with disabilities and supports inclusion and full participation in society. These aspects of the technologies align more closely with the social model. Both technologies embrace that treatment and rehabilitation are important while simultaneously creating an interface in the form of modified ride-on cars between children with disabilities and their environment as a means to reduce barriers and support social participation. Perhaps the social model of disability needs even further influence in rehabilitative practices surrounding self-directed mobility. Although commonly cited as at odds with rehabilitation frameworks, the ICF incorporates psychological and social (environmental) factors of the social model, and provides a framework to engage these models in a complimentary way as demonstrated in the current technical report (Wiat and Darrah, 2002; Gibson et al., 2009; Gibson and Teachman, 2012; McKeever et al., 2013).

There is a growing alternative culture within pediatric rehabilitation that highlights a social model of disability and embraces "effective mobility," defined as "locomotion that is functional, timely and energy-efficient" (Butler, 2009, p. 5). Effective mobility emphasizes a multi-modal view, suggesting that individuals, including young children, may need several options for

Throw Baby Throw Ride-on Car Technology

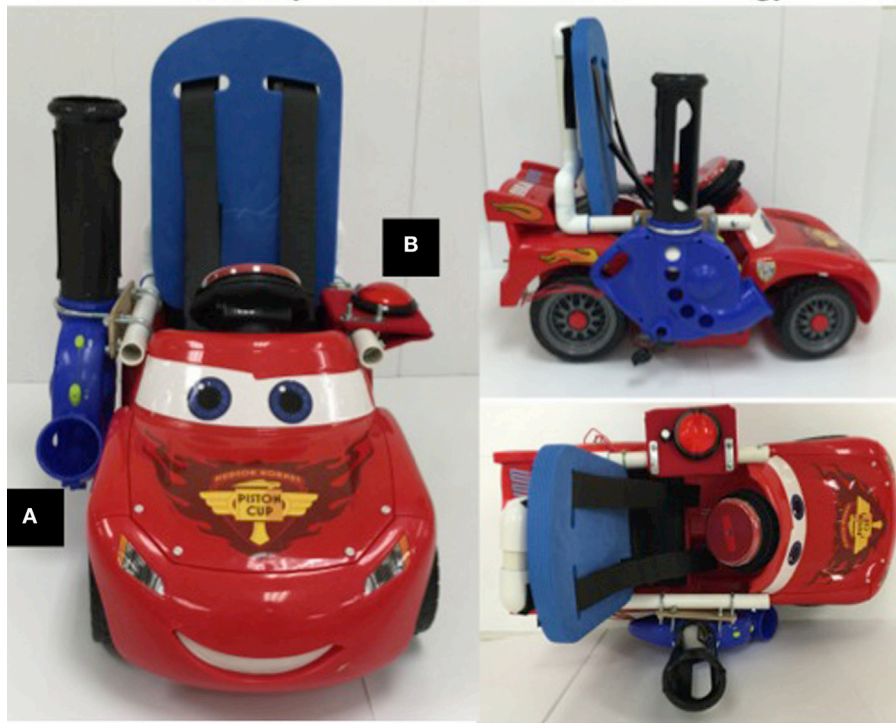


图7 | (A) Fisher-Price© 三重击棒球玩具投球机；(B) Adafruit© 街机按钮和电位器安装在自定义木质面板支架系统上。

残疾。然而，关于残疾模型本身及其影响的知识尚未转化为大多数普通公众的日常社区生活。残疾模型是一套关于残疾的原因、性质和治疗的信念。医学模型将残疾视为个体病理，通过基于偏离统计确定的标准的诊断标准来确定，以便进行治疗或治愈 (Olkin, 2002)。另一种观点是残疾的社会模型。在该模型中，残疾的“问题”主要是社会建构的，社会缺乏对人们差异的适应是导致残疾体验的原因 (Olkin, 2002)。因此，尽管在个体层面上可能存在功能障碍，但残疾的发生是因为社会的结构假设每个人都是健康的、没有残疾的，并符合文化构建的规范或理想 (Wendell)

本技术报告中提出的两种技术都强调通过提高身体技能来关注个体层面的障碍。这些技术方面更贴近医疗模式。然而，身体技能的培养被融入到儿童的驾驶玩具车、日常游戏、探索和参与体验中，这些体验被数百万儿童所享受，而无需

残疾。两种技术都为有残疾的儿童提供了一种参与类似童年经历的方式。技术的低成本、低技术含量和自行制作的特点促进了有残疾儿童的更高可及性，并支持社会中的包容性和充分参与。这些技术特点与社会模式更加紧密地契合。两种技术都认为治疗和康复很重要，同时通过改装的可驾驶玩具车在有残疾的儿童与其环境之间创建接口，作为减少障碍和支持社会参与的一种方式。也许残疾的社会模式在自我移动性相关的康复实践中需要更深入的影响。尽管常被认为与康复框架相矛盾，ICF 也纳入了社会模式中的心理和社会（环境）因素。

在儿童康复中，出现了一种日益增长的替代文化，它强调残疾的社会模型，并倡导“有效的移动性”，定义为“功能性、及时且节能的移动” (Butler, 2009, 第5页)。有效的移动性强调一种多模式的观点，表明包括幼儿在内的个体可能需要多种选择以

locomotion to provide opportunities for participation in daily life depending upon the environment and situation (Feldner et al., 2016). The multi-modal view of self-directed mobility more closely aligns with the social model of disability because it does not view a lack of independent locomotion as a disability, but rather embraces alternative ways that society may accommodate for people's differences by supporting access to powered mobility devices.

CONCLUSION

As a practical application of the shift in thinking to effective mobility, pediatric rehabilitation professionals can engage in more value-neutral modes of caregiver education, describing the benefits of self-directed mobility at all levels, whether generated *via* activation of a child's own body or *via* a mobility device, and emphasizing that *both* modes should occur as early as possible. Adjustments in service delivery can also include an explicit commitment to a paradigm of "right mode, right time" decision-making, in which all modes of self-directed mobility are considered equally valuable, dependent on environmental contexts and needs of the child. Doing so further reinforces the value of multi-modal self-directed mobility, and weakens a walking vs. wheelchair dichotomization by considering other forms of locomotion like bicycles, ride-on toys, the Segway, go-carts, and scooters (Feldner et al., 2016). Moreover, a team approach should be utilized to ensure that the child himself or herself is at the

center of these modal decision-making processes, at the earliest possible opportunity.

AUTHOR NOTES

There are currently no plans to pursue commercialization of either technology presented in this report. It is critical that any electrical and structural modifications to a ride-on car or toy-based pitching machine be fully tested and checked by an engineer with the appropriate expertise to determine safety. The technologies described in this report may or may not be appropriate for any individual child. It is important that families work with their child's clinical team in order to reach appropriate decisions of treatment on an individual basis.

AUTHOR CONTRIBUTIONS

SL, JG, WS, HF, and KB created the idea and were significantly involved in writing and reviewing manuscript drafts. BG, SR, MC, AW, and ZS were involved in the design, prototyping, and testing of the technologies, as well as reviewing manuscript drafts.

FUNDING

SL and JCG were supported by NIH NICHD NCMRR AWARD 5R21HD078708 (PI JCG). JCG was supported by NSF AWARD 1252876 (PI JCG).

REFERENCES

- Anderson, D. I., Campos, J. J., Witherington, D. C., Dahl, A., Rivera, M., He, M., et al. (2013). The role of locomotion in psychological development. *Front. Psychol.* 4:440. doi:10.3389/fpsyg.2013.00440
- Bogart, K. R. (2014). The role of disability self-concept in adaptation to congenital or acquired disability. *Rehabil. Psychol.* 59, 107–115. doi:10.1037/a0035800
- Bottos, M., and Gericke, C. (2003). Ambulatory capacity in cerebral palsy: prognostic criteria and consequences for intervention. *Dev. Med. Child Neurol.* 45, 786–790. doi:10.1017/S0012162203001452
- Butler, C. (2009). *Effective Mobility for Children with Motor Disabilities*. Available at: http://www.global-help.org/publications/books/help_effectivemobility.pdf
- Butler, C., Okamoto, G. A., and McKay, T. M. (1983). Powered mobility for very young disabled children. *Dev. Med. Child Neurol.* 25, 472–474. doi:10.1111/j.1469-8749.1983.tb13792.x
- Campos, J. J., Anderson, D. I., Barbu-Roth, M. A., Hubbard, E. M., Hertenstein, M. J., and Withering, D. (2000). Travel broadens the mind. *Infancy* 1, 149–219. doi:10.1207/S15327078IN0102_1
- Carter, E. W., Sisco, L. G., Chung, Y. C., and Stanton-Chapman, T. L. (2010). Peer interactions of students with intellectual disabilities and/or autism: a map of the intervention literature. *Res. Pract. Persons Sev. Disabil.* 35, 63–79. doi:10.2511/rpsd.35.3-4.63
- Chiulli, C., Corradi-Scalise, D., and Donatelli-Schultheiss, L. (1988). Powered mobility vehicles as aids in independent locomotion for young children. Suggestion from the field. *Phys. Ther.* 68, 997–999.
- Clark, J. E., and Metcalfe, J. S. (2002). "The mountain of motor development: a metaphor," in *Motor development: Research and review*, Vol. 2. eds J. E. Clark, and J. H. Humphrey (Reston, VA: National Association for Sport and Physical Education), 62–95.
- Cooper, R., Gottlieb, P., and Koeff, B. (1992). "A new low-cost approach to early powered mobility – the Coopercar," in *Proceedings of the Johns Hopkins Search for Computing Applications to Assist Persons with Disabilities*, Laurel, MD, 146–148.
- Cooper, R. A., Tolerico, M., Kaminski, B. A., Spaeth, D., Ding, D., and Cooper, R. (2008). Quantifying wheelchair activity of children: a pilot study. *Am. J. Phys. Med. Rehabil.* 87, 977–983. doi:10.1097/PHM.0b013e31818dfe74
- Daudji, A., Eby, S., Foo, T., Ladak, F., Sinclair, C., Landry, M. D., et al. (2011). Perceptions of disability among south Asian immigrant mothers of children with disabilities in Canada: implications for rehabilitation service delivery. *Disabil. Rehabil.* 33, 511–521. doi:10.3109/09638288.2010.498549
- Dunn, D. S., and Burcaw, S. (2013). Disability identity: exploring narrative accounts of disability. *Rehabil. Psychol.* 58, 148–157. doi:10.1037/a0031691
- Feldner, H. A., Logan, S. W., and Galloway, J. C. (2016). Why the time is right for a radical paradigm shift in early power mobility: the role of mobility technology devices, policy, and stakeholders. *Disabil. Rehabil. Assist. Technol.* 11, 89–102. doi:10.3109/17483107.2015.1079651
- Fidler, D. J. (2005). The emerging Down syndrome behavioral phenotype in early childhood. *Infants Young Child.* 18, 86–103. doi:10.1097/00001163-200504000-00003
- Galloway, J. C., Ryu, J. C., and Agrawal, S. K. (2008). Babies driving robots: self-generated mobility in very young infants. *J. Intell. Serv. Rob.* 1, 123–134. doi:10.1007/s11370-007-0011-2
- Gibson, B., Darrah, J., Cameron, D., Hashemi, G., Kingsnorth, S., Lepage, C., et al. (2009). Revisiting therapy assumptions in children's rehabilitation: clinical and research implications. *Disabil. Rehabil.* 31, 1446–1453. doi:10.1080/09638280802621390
- Gibson, B. E., Carnevale, F. A., and King, G. (2012). 'This is my way': reimagining disability, independence and interconnectedness of persons and assistive technologies. *Disabil. Rehabil.* 34, 1894–1899. doi:10.3109/09638288.2012.670040
- Gibson, B. E., and Teachman, G. (2012). Critical approaches in physical therapy research: investigating the symbolic value of walking. *Physiother. Theory Pract.* 28, 474–484. doi:10.3109/09593985.2012.676936
- Gibson, B. E., Teachman, G., Wright, V., Fehlings, D., Young, N. L., and McKeever, P. (2011). Children's and parents' beliefs regarding the value of walking: rehabilitation implications for children with cerebral palsy. *Child Care Health Dev.* 38, 61–69. doi:10.1111/j.1365-2214.2011.01271.x

行动能力为日常生活中的参与提供机会，这取决于环境和情境（Feldner 等人，2016）。自我主导移动的多模式观点与社会性残疾模型更为一致，因为它不将缺乏独立行动能力视为一种残疾，而是通过支持使用动力移动设备来接纳社会为人们差异提供便利的替代方式。

结论

作为将思维转向有效移动的一个实际应用，儿童康复专业人员可以参与更多价值中性方式的看护者教育，描述自我引导移动在各个层面的好处，无论是通过激活孩子自己的身体产生，还是通过移动设备产生，并强调这两种方式都应尽早进行。服务提供的调整还可以包括明确承诺采用“正确模式，正确时间”的决策范式，其中所有自我引导移动的模式都被认为同等有价值，这取决于环境背景和孩子需求。这样做进一步强化了多模式自我引导移动的价值，并通过考虑其他形式的移动方式（如自行车、儿童电动车、赛格威、卡丁车和滑板车）来削弱步行与轮椅的二分法（Feldner 等人，2016）。此外，一个团队的应用…

参考文献

Anderson, D. I., Campos, J. J., Witherington, D. C., Dahl, A., Rivera, M., He, M. 等人 (2013). 运动在心理发展中的作用。心理学前沿 4:440. doi:10.3389/fpsyg.2013.00440 Bogart, K. R. (2014). 残障自我概念在适应先天或后天残障中的作用。康复心理学 59, 107–115. doi:10.1037/a0035800 Bottos, M., 和 Geri cke, C. (2003). 脑性瘫痪患者的步行能力：预测标准及干预后果。发展医学与儿童神经学 45, 786–790. doi:10.1017/S0012162203001452 Butler, C. (2009). 运动功能障碍儿童的有效移动能力。可在以下网址获取：http://www.global-help.org/publications/books/help_effectivemobility.pdf Butler, C., Okamoto, G. A., 和 McKay, T. M. (1983). 非常年幼的残障儿童的动力移动能力。发展医学与儿童神经学 25, 472–474. doi:10.1111/j.1469-8749.1983.tb13792.x Campos, J. J., Anderson, D. I., Barbu-Roth, M. A., Hubbard,

在这些模式化决策过程的中心，尽早抓住可能的机会。

作者注

目前没有计划对本报告中提出的任何技术进行商业化。对乘坐式汽车或基于玩具的投球机进行的任何电气和结构改装，必须由具备相应专业知识的工程师进行全面测试和检查，以确定安全性。本报告中描述的技术可能适合，也可能不适合任何个体儿童。家庭应与孩子的临床团队合作，以便在个体基础上做出适当的治疗决策，这一点非常重要。

作者贡献

SL、JG、WS、HF 和 KB 提出了这个想法，并在手稿草稿的撰写和评审中发挥了重要作用。BG、SR、MC、AW 和 ZS 参与了技术的设计、原型制作和测试，以及手稿草稿的评审。

资金

SL 和 JCG 获得了美国国立卫生研究院 NICHD NCMRR 资助 5R21HD078708（首席研究员 JCG）。JCG 获得了美国国家自然科学基金会资助 1252876（首席研究员 JCG）。

库珀, R. A., 托勒里科, M., 卡明斯基, B. A., 斯帕斯, D., 丁, D., 和 库珀, R. (2008). 儿童轮椅活动量化：一项试点研究。美国物理医学与康复杂志, 87, 977–983. doi:10.1097/PHM.0b013e31818dfe74 Daudji, A., 埃比, S., 傅, T., 拉达克, F., 辛克莱, C., 兰德里, M. D., 等 (2011). 加拿大南亚移民中患有残疾儿童母亲对残疾的认知：康复服务提供的启示。残疾与康复, 33, 511–521. doi:10.3109/09638288.2010.498549 邓恩, D. S., 和 伯考, S. (2013). 残疾身份：探索残疾的叙事记录。康复心理学, 58, 148–157. doi:10.1037/a0031691 费尔德纳, H. A., 洛根, S. W., 和 加洛韦, J. C. (2016). 为什么现在是早期动力移动领域进行彻底范式转变的合适时机：移动技术设备、政策与利益相关者的作用。残疾康复辅助技术, 11, 89–102. doi:10.3109/17483107.2015.1079651 菲德勒, D. J.

- Haywood, K. M., and Getchell, N. (2009). *Lifespan Motor Development*, 5th Edn. Champaign, IL: Human Kinetics.
- Held, R., and Hein, A. (1963). Movement-produced stimulation in the development of visually guided behavior. *J. Comp. Physiol. Psychol.* 56, 872–876. doi:10.1037/h0040546
- Huang, H., and Galloway, J. C. (2012). Modified ride-on toy cars for early power mobility: a technical report. *Pediatr. Phys. Ther.* 24, 149–154. doi:10.1097/PEP.0b013e31824d73f9
- Jones, M. A., McEwen, I. R., and Neas, B. R. (2012). Effects of power wheelchairs on the development and function of young children with severe motor impairments. *Pediatr. Phys. Ther.* 24, 131–140. doi:10.1097/PEP.0b013e31824c5fde
- Law, M., Darrah, J., Pollock, N., Rosenbaum, P., Russell, D., Walter, S. D., et al. (2007). Focus on function – a randomized control trial comparing two rehabilitation interventions for young children with cerebral palsy. *BMC Pediatr.* 7. doi:10.1186/1471-2431-7-31
- Libertus, K., and Needham, A. (2010). Teach to reach: the effects of active vs. passive reaching experiences on action and perception. *Vision Res.* 50, 2750–2757. doi:10.1016/j.visres.2010.09.001
- Livingstone, R. (2010). A critical review of powered mobility assessment and training for children. *Disabil. Rehabil. Assist. Technol.* 5, 392–400. doi:10.3109/17483107.2010.496097
- Lobo, M. A., Harbourne, R. T., Dusing, S. C., and McCoy, S. W. (2013). Grounding early intervention: physical therapy cannot just be about motor skills anymore. *Phys. Ther.* 93, 94–103. doi:10.2522/ptj.20120158
- Logan, S. W., Feldner, H. A., Galloway, J. C., and Huang, H. H. (2016). Modified ride-on car use by children with complex medical needs. *Pediatr. Phys. Ther.* 28, 100–107. doi:10.1097/PEP.0000000000000210
- Logan, S. W., Feldner, H. A., Lobo, M. A., Winden, H. N., MacDonald, M., and Galloway, J. C. (2017). Power, up: exploration and play in a novel modified ride-on car for standing. *Pediatr. Phys. Ther.* 29, 30–37. doi:10.1097/PEP.0000000000000336
- Logan, S. W., Huang, H. H., Stahl, K., and Galloway, J. C. (2014). Modified ride-on car use for mobility and socialization: single-case study of an infant with Down syndrome. *Pediatr. Phys. Ther.* 26, 418–426. doi:10.1097/PEP.0000000000000070
- Logan, S. W., Robinson, L. E., Wilson, A. E., and Lucas, W. A. (2011). Getting the fundamentals of movement: a meta-analysis of the effectiveness of motor skill interventions in children. *Child Care Health Dev.* 38, 305–315. doi:10.1111/j.1365-2214.2011.01307.x
- Lynch, A., Ryu, J., Agrawal, S., and Galloway, J. C. (2009). Power mobility training for a 7-month-old infant with spina bifida. *Pediatr. Phys. Ther.* 21, 362–368. doi:10.1097/PEP.0b013e3181bf4e4c
- McKeever, P., Rossen, B. E., Scott, H., Robinson-Vincent, K., and Wright, V. (2013). The significance of uprightness: parents' reflections on children's responses to a hands-free walker for children. *Disabil. Soc.* 28, 380–392. doi:10.1080/09687599.2012.714259
- Nicholson, J., and Bonsall, M. (2002). Powered mobility for children under five years of age in England. *Br. J. Occup. Ther.* 65, 291–293. doi:10.1177/030802260206500608
- Olkin, R. (2002). Could you hold the door for me? Including disability in diversity. *Cultur. Divers. Ethnic Minor. Psychol.* 8, 130–137. doi:10.1037/1099-9809.8.2.130
- Ostensjø, S., Carlberg, E. B., and Vøllestad, N. K. (2005). The use and impact of assistive devices and other environmental modifications on everyday activities and care in young children with cerebral palsy. *Disabil. Rehabil.* 27, 849–861. doi:10.1080/09638280400018619
- Ragonesi, C. B., Chen, X., Agrawal, S., and Galloway, J. C. (2010). Power mobility and socialization in preschool: a case study of a child with cerebral palsy. *Pediatr. Phys. Ther.* 22, 322–329. doi:10.1097/PEP.0b013e3181eab240
- Robinson, L. E., Stodden, D. F., Barnett, L. M., Lopes, V. P., Logan, S. W., D'Hondt, E., et al. (2015). Motor competence and its effect on positive developmental trajectories of health. *Sports Med.* 45, 1273–1284. doi:10.1007/s40279-015-0351-6
- Rosen, L., Roos, J., Sabet, A., Vander Schaaf, P., Wonsettler, T., Arva, J., et al. (2009). RESNA position on the application of power wheelchairs for pediatric users. *Assist. Technol.* 21, 218. doi:10.1080/10400430903246076
- Schoenmakers, M. A., Uiterwaal, C. S., Gultmans, V. A., Gooskens, R. H., and Hadders, P. J. (2005). Determinants of functional independence and quality of life in children with spina bifida. *Clin. Rehabil.* 19, 677–685. doi:10.1191/0269215505cr8650a
- Smith, L. B., and Gasser, M. (2005). The development of embodied cognition: six lessons from babies. *Artif. Life* 11, 13–30. doi:10.1162/1064546053278973
- Staincliffe, S. (2003). Wheelchair services and providers: discriminating against disabled children? *Br. J. Ther. Rehabil.* 10, 151–159. doi:10.12968/bjtr.2003.10.4.13555
- Thelen, E., Schoner, G., Scheier, C., and Smith, L. B. (2001). The dynamics of embodiment: a field theory on infant preservative reaching. *Behav. Brain Sci.* 24, 1–86. doi:10.1017/S0140525X01003910
- Thelen, E., and Smith, L. B. (1994). *A Dynamic Systems Approach to the Development of Cognition and Action*. Cambridge, MA: MIT Press.
- Tudella, E., Pereira, K., Basso, R. P., and Svendsbergh, G. J. P. (2011). Description of the motor development of 3–12 month old infants with Down syndrome: the influence of the postural body position. *Res. Dev. Disabil.* 32, 1514–1520. doi:10.1016/j.ridd.2011.01.046
- Uchiyama, I., Campos, J. J., Frankel, C. B., Anderson, D. I., Witherington, D., Lejeune, L., et al. (2008). Locomotor experience affects self and emotion. *Dev. Psychol.* 44, 1225–1231. doi:10.1037/a0013224
- Ulrich, D. A., Ulrich, B. D., Angulo-Kinzler, R. M., and Yun, J. (2001). Treadmill training of infants with Down syndrome: evidence-based developmental outcomes. *Pediatrics* 108, E84. doi:10.1542/peds.108.5.e84
- Webster, A. A., and Carter, M. (2007). Social relationships and friendships of children with developmental disabilities: implications for inclusive settings. A systematic review. *J. Intellect. Dev. Disabil.* 32, 200–213. doi:10.1080/13668250701549443
- Wendell, S. (1996). *The Rejected Body: Feminist Philosophical Reflections on Disability*. New York, NY: Psychology Press.
- Wiat, L., and Darrah, J. (2002). Changing philosophical perspectives on the management of children with physical disabilities – their effect on the use of powered mobility. *Disabil. Rehabil.* 24, 492–498. doi:10.1080/09638280110105240
- Wiat, L., Darrah, J., Hollis, V., Cook, A., and May, L. (2004). Mothers' perceptions of their children's use of powered mobility. *Phys. Occup. Ther. Pediatr.* 24, 3–21. doi:10.1300/J006v24n04_02
- Wu, J., Looper, J., Ulrich, D. A., and Angulo-Barroso, R. M. (2010). Effects of various treadmill interventions on the development of joint kinematics in infants with Down syndrome. *Phys. Ther.* 90, 1265–1276. doi:10.2522/ptj.20090281
- Wuang, Y. P., Wang, C. C., Huang, M. H., and Su, C. Y. (2008). Profiles and cognitive predictors of motor functions among early school-age children with mild intellectual disabilities. *J. Intellect. Disabil. Res.* 52, 1048–1060. doi:10.1111/j.1365-2788.2008.01096.x
- World Health Organization. (2001). *International Classification of Functioning, Disability, and Health*. Geneva: WHO.

Conflict of Interest Statement: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Copyright © 2017 Logan, Feldner, Bogart, Goodwin, Ross, Catena, Whitesell, Sefton, Smart and Galloway. This is an open-access article distributed under the terms of the Creative Commons Attribution License (CC BY). The use, distribution or reproduction in other forums is permitted, provided the original author(s) or licensor are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

- Haywood, K. M., 和 Getchell, N. (2009)。终身运动发展, 第5版。伊利诺伊州尚佩恩: 人类动力学出版社。Held, R., 和 Hein, A. (1963)。运动产生的刺激在视觉指导行为发展中的作用。《比较生理心理学杂志》56, 872–876。doi:10.1037/h0040546 Huang, H., 和 Galloway, J. C. (2012)。用于早期动力移动的改装骑行玩具车: 一份技术报告。《儿科物理治疗》24, 149–154。doi:10.1097/PEP.0b013e31824d73f9 Jones, M. A., McEwen, I. R., 和 Neas, B. R. (2012)。动力轮椅对严重运动障碍幼儿发展与功能的影响。《儿科物理治疗》24, 131–140。doi:10.1097/PEP.0b013e31824c5fdc Law, M., Darrah, J., Pollock, N., Rosenbaum, P., Russell, D., Walter, S. D., 等 (2007)。关注功能——一项随机对照试验比较两种脑性瘫痪幼儿康复干预。《BMC儿科》7。doi:10.1186/1471-2431-7-31 Libertus, K., 和 N Rosen, L., Roos, J., Sabet, A., Vander Schaaf, P., Wonsettlar, T., Arva, J., 等 (2009)。RESNA 关于儿童使用动力轮椅的立场。辅助技术, 21, 218。doi:10.1080/10400430903246076 Schoenmakers, M. A., Uiterwaal, C. S., Gulmans, V. A., Gooskens, R. H., 和 Helden, P. J. (2005)。脊柱裂儿童功能独立性和生活质量的决定因素。临床康复, 19, 677–685。doi:10.1191/0269215505cr865oa Smith, L. B., 和 Gasser, M. (2005)。具身认知的发展: 来自婴儿的六个教训。人工生命, 11, 13–30。doi:10.1162/1064546053278973 Staincliffe, S. (2003)。轮椅服务与提供者: 是否歧视残疾儿童? 英国治疗与康复杂志, 10, 151–159。doi:10.12968/bjtr.2003.10.4.13555 Thelen, E., Schoner, G., Scheier, C., 和 Smith, L. B. (2001)。具身性的动态: 婴儿保守性伸手动作的场论。行为与脑科学

利益冲突声明: 作者声明本研究是在不存在任何可被视为潜在利益冲突的商业或财务关系的情况下进行的。

版权所有 © 2017 Logan, Feldner, Bogart, Goodwin, Ross, Catena, White sell, Sefton, Smart 和 Galloway。这是一篇开放获取文章, 根据知识共享署名许可协议 (CC BY) 进行分发。允许在其他平台上使用、分发或再版, 只要原作者或许可方得到署名, 并且引用该期刊的原始发表文章, 符合公认的学术规范。不允许任何不符合这些条款的使用、分发或再版。